CHAGEE

CHAGEE SINGAPORE

FORECAST THE PEAK, SHARE THE COUNTER

PREDICTIVE AI AND COUNTER AUTOMATION: WHAT THE GM DECIDES FOR THE BUSIEST HOUR

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN · OCTOBER 2026



SPEAKER NOTES · TITLE

Good morning. We are Group 3. Our company is Chagee, and we look at one place in its business: the counter of a Singapore outlet in its busiest hour.

Two kinds of technology meet there. Predictive AI forecasts how busy the hour will be; that is the AI in our project. Machines do the fixed steps; that is robotics and automation, not AI. Around them, people share the hand steps like a bucket brigade.

Our decision-maker throughout is the GM, Chagee Singapore's general manager. Timing: title and agenda together, about twenty seconds.

Agenda

01 **Chagee in Singapore**
How it earns and sells

02 **From farm to cup**
Where the supply chain is stable, and where it is not

03 **The data trail**
What every cup already records

04 **Our scope**
One area, one decision-maker

05 **Predict demand**
App searches, DBSCAN and a new-store pin

06 **Share the counter**
The bucket brigade, side by side

07 **Split the counter**
Which steps stay human: the GM's call

08 **Machine ladder**
What machines exist, what they cost, when to add them

09 **The GM's playbook**
Triggers, pilot and limits



SPEAKER NOTES · AGENDA

Eight short chapters. The first four set the scope; the last four are the forecast, the counter, the GM's split and the playbook. Click any chapter to jump there.

In Singapore, outlets run under one company, and every channel ends at one counter

One company and one data system: the counter, the app data and the call on who works the peak sit together (informal source, to confirm).

HOW A SINGAPORE CUP IS SOLD · STEP THROUGH

GROUP NET REVENUE, FY 2025 (SECONDARY SOURCE)



Franchise **88%** Company-owned **12%**, growing fastest

SG Singapore outlets run under one company and one data system (informal source, to confirm with Chagee)

In store	walk-in, counter or kiosk
Chagee app	order ahead, collect
Grab	third-party delivery

→ **One counter** every channel's cup is made here



44 outlets, ground-floor mall units by preference; kiosks in basements and MRT stations at entry in 2019

- **One operator, one data system**
  Singapore outlets run under one company and one data system, so a counter design or roster rule could be rolled out across outlets.
  A Chagee contact (informal), to confirm with Chagee
- **App-first**
  Most orders come through the app, with few kiosks in store. Every order and search is logged: the raw material for a forecast.
  Team observation, to confirm
- **Why Chagee**
  A fast-growing, playful brand with heavy app use: plenty of data, and room for AI to help the stores.

TAKEAWAY | Whatever the channel, every order ends at one counter. The peak hour is where an outlet wins or loses.

SPEAKER NOTES · CHAGEE IN SINGAPORE

Step through four points. One: at group level most revenue comes from franchising, about 88% in FY 2025; company-owned is about 12% and growing fastest. That split is from a secondary source.

Two: Singapore outlets run under one company and one data system. We have this informally from a Chagee contact, who told us the ownership form, owned, joint venture or franchise, does not matter for this question. We will confirm it with Chagee, and we make no claim about the ownership structure.

Three: 44 outlets, by the store directory in October. Chagee prefers ground-floor mall units.

Four: walk-in, app orders for collection, and Grab delivery all end at the same counter. The Chagee app is for ordering ahead and collecting; delivery goes through third-party apps, by the same informal source. That is why we look at the counter. About fifty seconds.

Supply is standard and planned ahead; the counter feels demand within minutes

Tea leaves travel one standard route. Only fresh milk and fruit are bought locally.

FOLLOW ONE CUP · STEP THROUGH



Source

Tea leaves from Yunnan through standard channels for every market

DATA CREATED

Shipments, lead times

Local: fresh milk, fruit

Warehouse

Separate central and regional warehouses keep storage and logistics costs down

DATA CREATED

Stock vs par, expiry

Outlet counter

Machines brew each cup from a scanned recipe; people do ice, toppings and hand-over

DATA CREATED

Cups by hour

Wait times

Last mile

Self-collect from the app, or delivery via third-party apps such as Grab; riders wait at the same counter

DATA CREATED

Orders by channel

Rider wait

Customer

Orders in the app, collects at the counter, comes back (or not)

DATA CREATED

App orders, drop-offs

- **Stable upstream**

  Standard sourcing, regional warehouses and forecast-led restocking: Chagee already plans this part.

  YF Chuhai; Chagee Form 20-F

- **Volatile downstream**

  Demand at a counter moves by the hour: lunch, a mall event, rain, a new drink launch.

- **So we stop at the counter**

  The open question is not the shipment. It is how many people the counter needs this hour, and how they work.

TAKEAWAY | Chagee already plans its stock. The open problem is matching people to the hourly peak.

SPEAKER NOTES · FROM FARM TO CUP

Press play and the cup travels farm to customer. Sourcing is standard; only fresh milk and fruit are local (YF Chuhai). Chagee runs central and regional warehouses (Form 20-F) and already plans restocking from forecasts.

The volatile part is the counter, the red chips. Under time pressure, summarise this slide in one sentence: supply is planned, the counter is not. About forty seconds.

Every cup is already a timestamped record: the inputs for a forecast exist today

Four data sources and six decisions they could feed. We follow the four that start in the app and end at the counter.

WHICH SOURCE FEEDS WHICH DECISION · STEP THROUGH

	Store and production	App and members	Delivery	Supply and context
Demand forecasts	Cups per hour by drink and channel	Member repeat demand	Orders per hour by platform	Weather, holidays, launches
Staffing and prep	Wait times, machine logs	Pickup no-shows	Hand-over peaks	Labour hours vs cups per hour
Replenishment	Drink mix per outlet	Launch trial rates	Group-order toppings	Ingredient use, expiry, par
Queue and capacity	Wait time, footfall vs sales	Store-select drop-offs	Rider wait at counter	Stock-outs at peak
Site selection	Sales density per outlet	"No store nearby", home and work	Distance and fee by postcode	Mall footfall, competitors
Menu and promos	Drink share, customisation	Trial and repeat, voucher uplift	Delivery basket mix	Collab calendar

White chips mark the core input; grey text refines it. Red rows: our scope.

- **No new sensors needed**
  Point-of-sale, app and machine logs already exist. The work is joining them by outlet and hour.
- **The app is the link**
  Only the app ties a cup to a person. That needs consent and care under Singapore's data protection law (PDPA).
- **What we take forward**
  Demand forecast, then staffing, then queue capacity: one chain that ends with the GM's counter and roster.

TAKEAWAY | The data for an hourly forecast is already collected. What is missing is turning it into a roster.

SPEAKER NOTES · THE DATA TRAIL

Each step lights up one decision and the data that feeds it. Skip to the last step under time pressure: our scope is the three counter rows: forecast, staffing, queue.

This list is our reading of what Chagee's systems record, based on public material and our industry conversation, not a Chagee data dictionary. About fifty seconds.

One area, the in-store peak, and one decision-maker: the GM

Ideation gave us three improvement areas. The counter is where AI and the course's operations tools meet.

FROM THREE IDEAS TO ONE · STEP THROUGH

FUTURE WORK

Online retailing

Steer orders between app pickup and Grab by hour.

Needs platform fees and price tests we do not have.

FORECAST ONLY

New stores

Forecast a new outlet from app searches (slide 05).

Rent, fit-out and payback stay future work.

OUR FOCUS

In-store operations

Forecast the peak, set the roster, share the counter.

Data exists, the decision is daily; Topic 3 forecasting, Topic 5 bucket brigades.

AI · PREDICTIVE

Forecast cups per hour from history, calendar and app orders.

AUTOMATION, NOT AI

Machines do the fixed steps, such as dispensing tea from a scanned recipe.

OPERATIONS TOOLS, NOT AI

Bucket brigade and cross-training: how people share the hand steps.

The GM

General manager, Chagee Singapore

DECIDES	Which counter steps go to robots, how many people work the peak, and how they share it
WHEN	Before a store opens, and when rosters are set
USES	The forecast and the counter simulation
TODAY	To confirm with Chagee: how rosters are set now

TAKEAWAY | Predictive AI says how many people the peak needs; the GM decides which steps robots do; the bucket brigade decides who does what.

SPEAKER NOTES · OUR SCOPE

Our ideation produced three areas. We park online retailing as future work, forecast new stores from app data, and focus on the in-store peak.

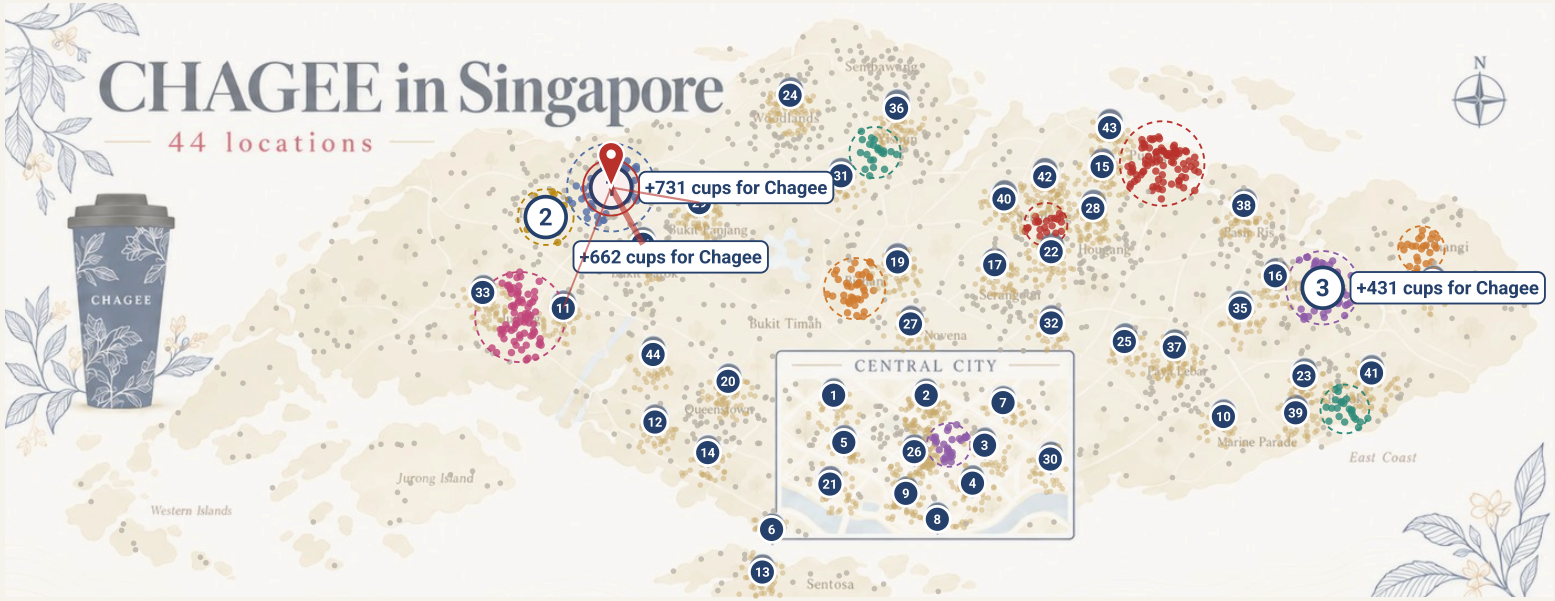
We are careful about what counts as AI: only the forecast. The machines are robotics and automation. Bucket brigade and cross-training are operations tools from Topic 5, order fulfilment, but they decide whether a lean roster actually works.

The decision-maker is the GM; outlet teams run the roster day to day. We do not yet know how Chagee sets rosters today; that is a question for our industry contact. About fifty seconds.

Where would outlet 45 add the most cups for Chagee?

Not what a new store sells, but what it adds once you subtract the cups it takes from nearby outlets.

NEW-STORE FORECAST



SUGGESTED SITE 1 · 2.7 KM FROM 34 · WEST MALL

Strong site: mostly new cups

+731 cups a day for Chagee

The new store sells 837 a day (1.5× an average outlet). 106 of those move from outlets nearby:

34 · West Mall, 2.7 km	**−77**
29 · Bukit Panjang Plaza, 3.3 km	**−28**

Peak hour 13:00: **92 cups** → **2 people** on today's counter (slides 06–07).

Synthetic app data on the team's map; outlet positions approximate (the central inset is enlarged). Not a trained model: the point is the method.

TAKEAWAY | Judge a site by what it adds to Chagee, not what it sells. The top cluster, near Choa Chu Kang, adds about 731 cups a day; a store beside an existing outlet would sell plenty, but about 80% of it is moved from that outlet.

SPEAKER NOTES · PREDICT DEMAND FOR A NEW STORE

Chagee Singapore runs on its app, with few kiosks, so every search is a demand signal, including searches where no store is close. That makes the app the forecasting input for a new store, before it has any sales.

Step one: synthetic app sessions on our map. Gold are within walking distance of an outlet; red found no store nearby. Step two: DBSCAN, unsupervised machine learning, groups red searches that are dense; scattered ones are noise. Step three: cluster centres are candidate sites.

Step four: click the map to test a site. The card leads with the gain for Chagee, which is the store's sales minus the cups it takes from nearby outlets, and gives a verdict: strong, mixed, or mostly moving cups. Try a pin next to Tampines West CC: it sells, but most of it moves from the two Tampines outlets nearby. Step five: the forecast's peak hour gives the opening roster for the counter on the next slides.

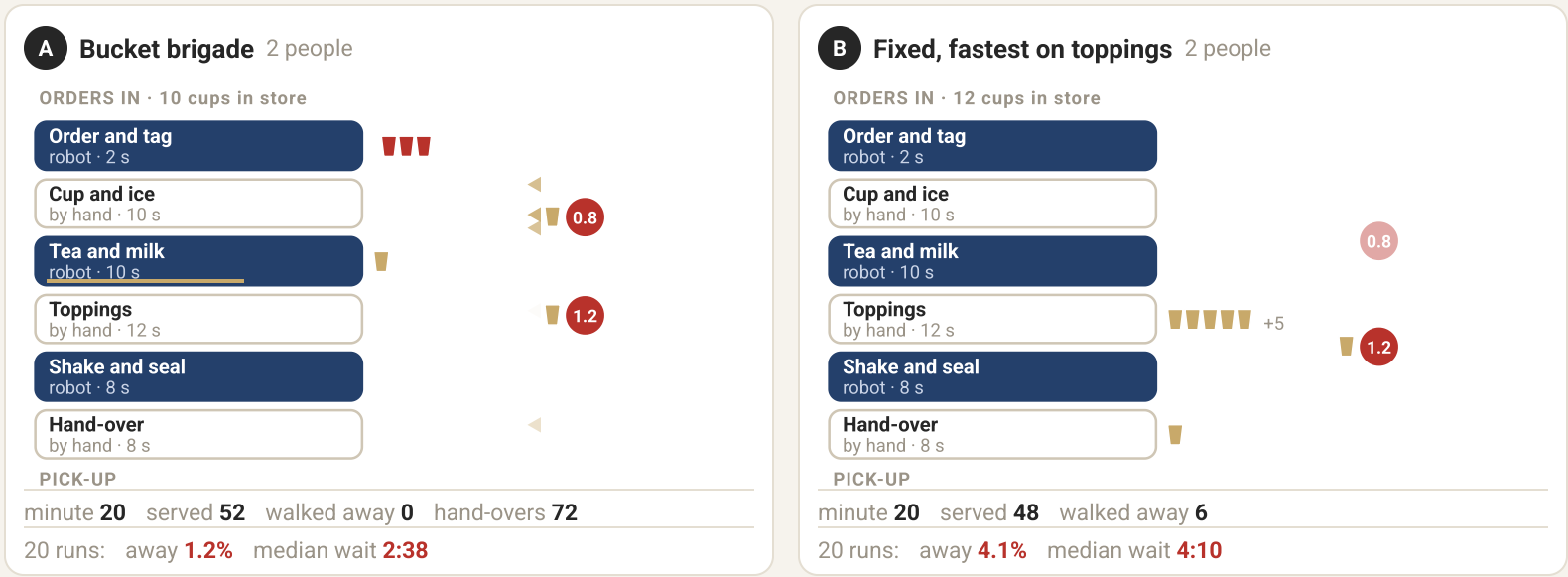
If asked how DBSCAN works, press How it works: it replays the algorithm one search at a time on a toy set. The appendix lists every simulated number.

The three sliders are the only inputs: cluster radius, minimum searches per cluster, and how far people will walk. The data is synthetic; with Chagee's real app logs, the same steps would run unchanged. About one and a half minutes.

With a lean roster, a bucket brigade keeps cups moving without knowing who is fastest

Today's counter, two ways of sharing the hand steps, side by side. Pick what each side runs.

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES



Red circle = a person (number = speed) · gold ring = held back · faded = walking · ▲ = hand-over · navy = robot · red cup = waiting for a person

ALL THREE RULES, MEAN OF 20 RUNS

EVENT DAY, 2 PEOPLE	WALKED AWAY	MEDIAN WAIT
Bucket brigade	**1.2%**	**2:38**
Fixed, fastest on toppings	4.1%	4:10
Fixed, slowest on toppings	34.9%	9:47

At the people on side A. A customer leaves if the wait would pass 5 min. With spare people, a well-assigned fixed line is as good or a little faster.

Why lean is where it counts

Work flows to whoever is free, so the line balances itself (Topic 5). Fixed stations are only as good as the guess of who is fast.

TAKEAWAY | At the forecast's roster on an event day, the brigade loses about 1.2% of orders; fixed stations lose 4.1% to 35%, depending on who is on toppings.

SPEAKER NOTES · SHARE THE COUNTER

Same counter, same customers, two sharing rules side by side. Robots, the navy boxes, take the order, brew tea and milk, and shake and seal. People do cup and ice, toppings and hand-over.

Left, a bucket brigade: each person carries a cup forward; the last hands it over, walks back and takes over a colleague's cup. Right, fixed stations: each person owns a station.

Press play on an event day. Then set the right side to "slowest on toppings": the queue builds at toppings and customers walk away. The brigade never needed to know who was fast.

Be honest about the other case: add people and fixed stations catch up, or are a little faster. About one minute.

Which steps stay human? Robots buy a shorter wait; on a normal day, people are still cheaper

Choose a split for each side, or click any station to hand it to a robot or a person, then run the peak hour.

PEAK HOUR · SAME CUSTOMERS ON BOTH SIDES

A Today's counter 2 people, brigade

ORDERS IN · 10 cups in store

- Order and tag — robot · 2 s
- Cup and ice — by hand · 10 s
- Tea and milk — robot · 10 s
- Toppings — by hand · 12 s
- Shake and seal — robot · 8 s
- Hand-over — by hand · 8 s

PICK-UP



minute **20** served **52** walked away **0** hand-overs **72**

20 runs: away **1.2%** wait **2:38** **S$402**/day

B Machine-made drink (scenario) 2 people, fixed posts

ORDERS IN · 0 cups in store

- Order and tag — by hand · 5 s
- Cup and ice — robot · 8 s
- Tea and milk — robot · 10 s
- Toppings — robot · 15 s
- Shake and seal — robot · 8 s
- Hand-over — by hand · 8 s

PICK-UP



minute **20** served **57** walked away **0**

20 runs: away **0.0%** wait **1:06** **S$465**/day

Navy = robot, white = by hand (click to switch) · red circle = a person · faded = walking · red cup = waiting for a person.

ALL FOUR SPLITS, MEAN OF 20 RUNS

EVENT DAY	PEOPLE	WALKED AWAY	WAIT	COST/DAY
Today's counter ✓	**2**	**1.2%**	**2:38**	**S$402**
Machine-made drink (scenario)	2 fixed	0.0%	1:06	S$465
All by hand	4	0.0%	1:25	S$422
All robots	–	0.0%	0:58	S$562

✓ cheapest within the limit · fastest: All robots

Each split gets the fewest people, from the forecast roster up, and the better sharing rule that keep walk-aways within 2%. Cost per day: people hours at the wage plus robots at their monthly cost.

Normal day, cost a day: All by hand S$361 (cheapest) · Today's counter S$372 · Machine-made drink (scenario) S$435 · All robots S$562. Fastest: All robots (0:52); today's counter is slower than all by hand (2:09 vs 1:16). Four splits only; a single machine can undercut all by hand (slide 08).

TAKEAWAY | On an event day, Today's counter is the cheapest split within the walk-away limit, at about S$402 a day. Machine-made drink (scenario) needs 2 people at fixed posts and costs about S$64 more, for a median wait of 1:06.

SPEAKER NOTES · SPLIT THE COUNTER

This is the GM's decision: which steps of the counter stay human and which go to robots. Side B, machine-made drink, is our scenario, not a claim about Chagee: machines make the drink while people tag the order and hand it over. Chagee's own equipment cost is not public; its Singapore relaunch used semi-automated equipment, according to Vulcan Post.

Click any station to switch it. Each side gets its forecast roster; change people with plus and minus. The table on the right is the simulation run for all four splits.

What it shows, honestly: the table is the event day; the line under it is the normal day, at about 556 cups. On a normal day all by hand is the cheapest of these four splits and all robots the fastest, so robots buy a shorter wait, not a lower cost; today's split is in fact slower than all by hand that day. On an event day, today's split is cheapest within the limit. The machine-made drink scenario needs a second person at the peak, because one person cannot tag and hand over at opposite ends of the counter.

These are four splits only. The next slide shows that a single machine can undercut all by hand, because it lets the roster drop a person off-peak. About one and a half minutes.

What machines exist, what they cost, and when to add them

To our knowledge, based on public sources; vendor list prices, not the cost to implement. Chagee's own equipment cost is not public; its Singapore relaunch used semi-automated equipment (Vulcan Post).

THE LADDER · NAVY = STEPS ON MACHINES

RUNG · EXAMPLE	VENDOR LIST PRICE · SOURCE	STEPS ON MACHINES	BEATS HAND, CUPS/DAY
1 **Semi-automation** Gong Cha Singapore	*Not disclosed* PRESS	Tag Seal	278–555
2 **Beverage platform** Gong cha Super Wu, Feb 2026	*Not disclosed* COMPANY CLAIM	Tag Tea Seal	278–693, 833
3 **Milk-tea machine** Amacha, one small shop	S$10,000 to buy PRESS	Tag Tea Seal	278–693, 833
◯ **Today's split** Our scenario (slides 06 and 07), not Chagee's disclosed setup	Model: S$1,909/mo per station unit OUR ASSUMPTION	Tag Tea Seal	Not in this range
4 **Robot arm** Richtech ADAM, makes boba	*Not disclosed, rental only* VENDOR PAGE	Tag Cup Tea Top Seal	No price to test
5 **Robot barista** Oli; coffee-first, no tea listed	US$4,990/mo lease, US$59,000 buy VENDOR PAGE	Tag Cup Tea Top Seal	833 at lease

Green = beats all by hand, across our per-outlet range of 278 to 833 cups a day; red line = your outlet. The model's S$1,909 a month is per station unit (our reading: a semi-automated station); robot prices are per setup. Semi-automation and the beverage platform are priced at that S$1,909 because their real prices are not disclosed.

Your outlet: 556 cups a day, wage S$15.06 an hour, S$1,909 a month per station unit (the model's defaults; sliders on screen).

A MONTH	SAVED	AGAINST ALL BY HAND
Semi-automation	S$3,614	**Break-even: setup under S$3,614/mo** Model: 2 × S$1,909, S$203 more
Beverage platform	S$3,614	**Break-even: setup under S$3,614/mo** Model: 1 × S$1,909, beats by S$1,706
Milk-tea machine	S$3,614	**Beats by S$1,706** Model: 1 × S$1,909. Simple payback, if the machine lets the roster drop one person-shift: 2.8 months on S$10,000
Today's split	S$5,422	**S$305 more** Model: 3 × S$1,909
Robot arm	S$5,422	**Break-even: setup under S$5,422/mo** No public price
Robot barista	S$5,422	**Oli lease S$6,350: S$928 more** Buy US$59,000: simple payback 13.8 months

Why a single machine pays: it saves 8 person-hours a day (S$3,614 a month), because in 8 off-peak hours the roster drops from 2 people to 1. Today's split saves 12 hours a day but pays for 3 station units. **Model assumption:** the minimum crew is 2 by hand and 1 with machines, and it drives this result; robot rungs save no more than today's split until about 757 cups a day. Station speeds are the model's, not vendor claims. Simple payback = list price ÷ labour saved a month; no installation, maintenance or financing.

TAKEAWAY | A single machine that lets the roster drop one person for 8 off-peak hours pays at 556 cups a day under the model; today's 3-station split does not (S$305 a month more than all by hand). Rosters step in whole people, so this flips with volume and wage.

SPEAKER NOTES · MACHINE LADDER

This is robotics and automation, not AI; the AI in our project is the forecast that sets the roster. The ladder runs from semi-automation, kiosks and sealers, up to a robot barista. Navy chips are the counter steps each rung puts on machines. The dashed row is our own scenario, today's split from slide 07, so the ladder does not start from an all-human counter.

Prices are vendor list prices from public sources, not the cost to implement. Gong cha and Richtech do not publish prices, so we show the cost a setup would need to stay under to break even. The robot barista is coffee-first; no tea drinks are listed for Oli.

On the right, set your outlet's cups a day, the wage and the machine rate. Why a single machine pays and today's split does not: one machine lets the roster drop from 2 people to 1 in 8 off-peak hours, about one person-shift a day, at the cost of one station unit. Today's split saves 12 person-hours but pays for 3 units, so it ends up a little dearer than all by hand. That minimum crew, 2 by hand and 1 with machines, is our model's assumption, and rosters step in whole people, so the answer flips with volume and wage.

Payback is simple payback, if the machine lets the roster drop one person-shift; it leaves out installation, maintenance and financing. Semi-automation and the beverage platform have no published price, so they are priced at the model's per-unit rate. Chagee's own equipment cost is not public. About one minute.

Choose the split, roster to the forecast, and share the work to suit the split

Three moves for the GM, each with a trigger, then a pilot to test them.

WHAT THE FORECAST CHANGES, PEAK HOUR, TODAY'S COUNTER

DAY	FORECAST PEAK, CUPS PER HOUR	PEOPLE AT THE PEAK	WALK-AWAYS, FORECAST ROSTER AND BRIGADE	WALK-AWAYS, NORMAL-DAY ROSTER
Normal day	61	**1**	0.9%	0.9%
Busy day	92	**2**	0.0%	**20.8%**
Event day	153	**2**	1.2%	**53.8%**

1 Choose the split

Robots where speed matters more than cost; people where the work varies: tagging and hand-over.

2 Roster to the forecast

On today's counter, one more person for about every 83 forecast cups in the peak hour.

3 Share to suit the split

A brigade when hand steps sit along the line; fixed posts when machines make the drink.

- **Next step: pilot in 3 outlets**

Shadow	Roster	Roll out
forecast only	staff to it	if KPIs hold

KPIs: cups per labour hour, wait time, walk-aways at the peak. Compare forecast and actual cups weekly.

- **Limits we know**

All figures are illustrative assumptions, not Chagee data. Robot speeds and costs are assumed; breakdowns and breaks are not modelled.

- **To confirm with Chagee**

How rosters are set today, robot costs per station, and whether outlets log wait times and walk-aways.

TAKEAWAY | Predictive AI sets how many people; the split sets what robots do; the sharing rule sets who does what. Next, test it in 3 outlets.

SPEAKER NOTES · THE GM'S PLAYBOOK

Three moves for the GM: choose the split, roster to the forecast, and share the work to suit the split. Then a pilot in 3 outlets: shadow, roster, roll out, judged on cups per labour hour, wait time and walk-aways.

Close on the limits: these are our assumptions, not Chagee data. New-store openings, the next part of our work, will use the same simulator before a store opens. About one minute, then questions.

THANK YOU

QUESTIONS AND DISCUSSION

OPIM639 · GROUP 3 · KEZIAH, JIN, AYUMI, REVAN

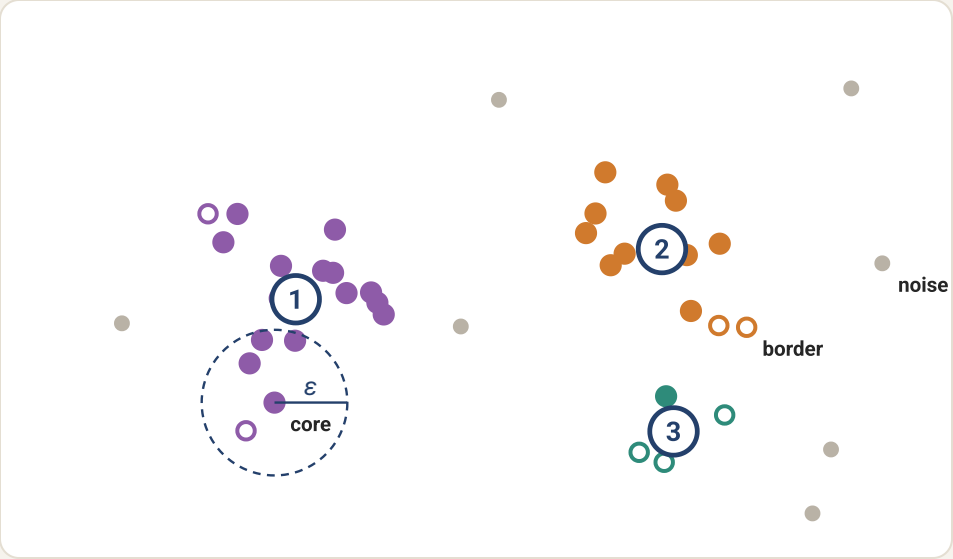


SPEAKER NOTES · THANK YOU

Likely questions: is the brigade always better? No; with spare staff fixed stations are as good. Is the forecast real? No, it is an illustration of the chain, built on stated assumptions.

DBSCAN finds where searches bunch up; a proximity model turns a pin into cups

DBSCAN IN FOUR MOVES · TOY EXAMPLE



1. **Radius ε.** Two searches are neighbours if they are within ε of each other (the cluster-radius slider).
2. **Core.** A search with at least the minimum number of searches within ε (counting itself) is a core point.
3. **Grow.** A cluster spreads from core to core. Searches it reaches that are not core join as border points.
4. **Noise.** Searches no cluster reaches. Unlike k-means, nobody picks the number of clusters, and odd shapes are fine.

Unsupervised: no labels or past openings are needed. Ester, Kriegel, Sander and Xu (1996). On slide 05 it runs only on searches with no outlet within walking distance; each cluster's centre is a candidate site.

THE SIMULATED NUMBERS, IN ORDER · AT THE SLIDER DEFAULTS

1	**App searches**	Synthetic dots on land: 15% spread evenly, 35% around the outlets' malls, the rest around 20 towns	**3,000 dots**
2	**No store nearby**	The nearest outlet is further than the walking distance	**1 km · 46% of dots**
3	**DBSCAN**	Clusters the no-store-nearby dots	**ε 0.6 km · min 15 · 11 clusters**
4	**Pull of a store**	Fades with distance d: exp(−½ (d/L)²), with the length L set by the walking distance	**L = 1 km**
5	**Who buys where**	A dot buys at a store in proportion to its pull, against the pull of not buying (a store next door = one)	**not buying 0.3**
6	**Calibration**	Each dot's weight is set so the average outlet sells the deck's cups a day (cups a year ÷ open days)	**556 cups a day**
7	**Score a pin**	Cups for Chagee = what the new store sells − what it moves from nearby outlets	**top: 837 − 106 = +731**
8	**Verdict**	Share of the new store's cups that are new to Chagee	**strong ≥ 60% · weak < 40%**
9	**Opening roster**	Peak-hour cups ÷ cups one person handles, with the safety margin	**one per 83 cups**

All synthetic: the dots, towns and weights are assumptions on the team's map (not to scale), not Chagee app logs or population data. With real app logs, steps one and two change; the rest runs unchanged.

SPEAKER NOTES · HOW THE SITE FORECAST WORKS

Backup for questions on slide 05. Left: DBSCAN in four moves on a toy set. A search is a core point if enough searches sit within the radius; clusters grow from core to core; border points join; the rest is noise. We do not choose the number of clusters.

Right: every simulated number, in the order the model uses it. The forecast is only as good as these assumptions; with Chagee's real app logs, the first two rows would be replaced by data and the rest would run as is.

APPENDIX · MACHINE LADDER, TIERS AND SOURCES (SLIDE 08)

Every rung, every price as published, and where it comes from

To our knowledge, based on public sources. Robotics and automation, not AI. A gap is "not disclosed", never a guess.

RUNG	EXAMPLE	AUTOMATES · STEPS ON MACHINES (OUR MAPPING)	VENDOR LIST PRICE	CLAIMS (NOT MODEL INPUTS)	RELIABILITY	SOURCE
1 **Semi-automation**	Gong Cha Singapore	Sugar dispensers, cup sealers, self-order kiosks. **Order and tag, Shake and seal**. Kiosk = order and tag, sealer = shake and seal. Sugar dispensing sits inside tea and milk, so it is not modelled.	*Not disclosed*	Over 20% productivity from self-ordering; 5% manpower saving (no baseline given)	PRESS	https://www.ntuc.org.sg/uportal/news/Technology-Brewing-Up-the-Bubble-Tea-Business/
2 **Beverage platform**	Gong cha Super Wu (Feb 2026)	Dispenses tea, sweeteners and syrups; brewing and shaking stay manual. **Tea and milk**	*Not disclosed*	Up to 65% higher peak productivity	COMPANY CLAIM	https://marketech-apac.com/gong-cha-launches-gong-cha-2-0-to-modernise-global-operations-and-store-design/
3 **Single automated milk-tea machine**	Amacha (one small shop, owner interview)	Makes the milk tea itself. **Tea and milk**	S$10,000 to buy	The shop's total opening cost was S$209,000 (not a machine cost)	PRESS	https://www.malaymail.com/news/life/2026/10/05/opening-a-milk-tea-shop-in-singapore-can-rack-up-to-s209000-even-before-the-first-cup-is-poured/237715
–	**Our scenario: today's split**	**Order and tag, Tea and milk, Shake and seal**. A scenario of ours (slides 06 and 07), not Chagee's disclosed setup	Model: S$1,909 a month per station unit		OUR ASSUMPTION	assumptions.json
4 **Robot arm that makes boba**	Richtech ADAM	Cup, ice, tea, toppings and sealing by one robot arm. **Cup and ice, Tea and milk, Toppings, Shake and seal**	*Not disclosed*. Rental only; price not published	–	VENDOR PAGE	https://richtechrobotics.com/solutions/adam
5 **Robot barista**	Oli	A whole drink station in one unit. **Cup and ice, Tea and milk, Toppings, Shake and seal**. Coffee-first. No tea drinks listed for Oli or Artly.	US$59,000 (≈ S$75,078) to buy; US$4,990 (≈ S$6,350) a month lease	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Artly		US$68,000–88,000 (≈ S$86,530–111,980) to buy; US$2,500–4,000 (≈ S$3,181–5,090) a month lease; US$2,500–4,000 a month software Oli's estimate of a competitor. The figures as given do not say whether the lease replaces or adds to the purchase, so no verdict is computed.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
	Cafe X		US$250,000 (≈ S$318,125) to buy Oli's estimate of a competitor's price.	–	VENDOR PAGE	https://olirobotics.com/best-robot-barista
–	**Chagee Singapore**	Chagee's own equipment cost is not public. Its Singapore relaunch used semi-automated equipment.	*Not public*	–	PRESS	https://vulcanpost.com/872176/chagee-singapore-comeback-after-sudden-exit/

Vendor list prices, not cost to implement: installation, training, maintenance, power, space and import are not included; USD converts at S$1.2725 per US$. **Simple payback** = list price ÷ labour saved a month, with no maintenance or financing. **The model's S$1,909 a month** is per station unit: our reading: one semi-automated station unit (for example a dispenser or a cup sealer), amortised with upkeep. Not a robot barista lease; robot prices in machine_tiers.json are per setup. **Vendor speed and throughput claims** are listed, never used: every rung runs at the model's station speeds. **Steps on machines** is our mapping. Artly and Cafe X prices are Oli's estimates of competitors.

SPEAKER NOTES · MACHINE LADDER: TIERS AND SOURCES

Backup for slide 08. Every price is as the source publishes it; where a source gives no price we say not disclosed. Claims such as Gong cha's productivity figures are listed for context and are not inputs to our model. Soft copy only.

Where every number comes from

FACTS QUOTED

Outlets in Singapore: 44. Store count used on the storyboard deck. The doc also cites 46 (VnExpress, earlier); the later directory count is used. *[L-sg-directory]*

Franchise share of FY2025 net revenue: 0.88. Secondary source; flagged wherever shown. *[L-koalagains]*

Company-owned share of FY2025 net revenue: 0.12. Secondary source; flagged wherever shown. *[L-koalagains]*

SOURCES

L-sg-directory Chagee Singapore store directory, mapped by the team (44 locations, checked 15 Oct 2026); Mothership, Sep 2026, https://mothership.sg/2026/09/chagee-more-heartland-stores-singapore-eugene-lee/ gives the same count. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-koalagains KoalaGains, 'CHA business and moat', https://koalagains.com/stocks/NASDAQ/CHA/business-and-moat (SECONDARY - flag when used): FY2025 net revenue about 88% from the franchise (master franchise) model and about 12% from company-owned stores, the latter growing fastest. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-yfchuhai YF Chuhai, https://www.yfchuhai.com/article/9751.html: Chagee's leaf supply chain is in China; it sources only two ingredients locally, fresh milk and fruit. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-sec20f Chagee Holdings Form 20-F for FY2025, SEC EDGAR, https://www.sec.gov/Archives/edgar/data/2013649/000110465926050766/cha-20251231x20f.htm: separate centralized and regional warehouses to lower storage and logistics costs. Taken from the team's ideation doc (OPIM 639_ Ideation Process), not re-checked for this deck.

L-unsw UNSW BusinessThink, Oct 2024: Chagee semi-automation, baristas scan a QR code and the machine dispenses the formulation. (brief.md section 9)

I-chagee-contact INFORMAL - a Chagee contact (Oct 2026). Three points only: whether Singapore outlets are owned, a joint venture or franchised does not matter for this work; Singapore outlets run under one company and one data system; all delivery is via third-party apps. Confirm with Chagee before citing.

I-sg-groundfloor INFORMAL - quote in the team's ideation doc (Chagee Singapore's Lee: 'Whenever you see a Chagee, it's always on the first floor'; kiosk-style outlets in basements and MRT stations at entry in 2019). The article is not named in the doc; confirm the source before quoting.

I-team-app-first INFORMAL - team observation in the ideation doc: Chagee Singapore sells mainly through its app, with minimal order kiosks in outlets. Confirm before citing.

L-ester1996 Ester, M., Kriegel, H.-P., Sander, J. and Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), 226-231. The DBSCAN algorithm used on slide 05.

ASSUMPTIONS (ASSUMPTIONS.JSON, ILLUSTRATIVE)

Cups per outlet per year: 200000 cups/store/year

Share of the day's cups in each open hour: 0.04, 0.06, 0.1, 0.11, 0.08, 0.08, 0.09, 0.1, 0.1, 0.09, 0.08, 0.07 share of daily cups per open hour (10:00 to 22:00), sums to 1

Busy and event day multipliers: none 1, moderate 1.5, large 2.5 x forecast arrival rate

Busiest hours the multiplier lands on: 2 busiest forecast hours that get the surge

Seconds per cup by hand, per station: ticket 5, cup ice 10, tea milk 15, topping finish 12, shake seal 10, handover 8 seconds of one average worker per cup

Seconds per cup by robot, per station: ticket 2, cup ice 8, tea milk 10, topping finish 15, shake seal 8, handover 6 seconds per cup per machine unit

Stations the robots do today: ticket, tea milk, shake seal

Spread of worker speeds: 0.2 +/- share around average speed

Minute-to-minute variation in pace: 0.25 coefficient of variation, minute to minute

Share of new staff: 0.25 share of roster in first months

New staff speed: 0.7 x average speed

Coordination loss, people around robots: 0.9 multiplier on human-pool throughput in the mixed setup

Minimum people on the counter: all human 2, mixed 1, all machine 0 workers per open hour

Roster safety margin over the forecast: 1.2 x forecast peak rate

Labour cost per hour worked (SGD): 15.06 SGD per hour worked, fully loaded (statutory costs)

Robot cost per station unit per month (SGD): 1908.75 SGD per station machine unit per month

Attendant hours, all-robot counter: 1 workers per open hour

Walk-away limit for the GM's split table: 0.02 share of peak-hour orders that walk away

Synthetic app-intent dots (slide 05): 3000 synthetic app-intent dots on the map

Walking distance to collect: 1 km a customer will walk to collect

Distance decay, share of the walking distance: 1 distance decay length as a share of walk_km

Pull of not buying: 0.3 pull of not buying, against a store next door (= 1)

DBSCAN cluster radius: 0.6 km

DBSCAN minimum searches: 15 dots within eps to form a cluster core

Map scale: 0.033 km per map pixel

Wait at which a customer walks away: 5 minutes

Walking-back speed (Week 5 units): 6 lines per hour

CLAIMS WE DID NOT MAKE

No Chagee sales, staffing or wait-time data is used; the forecast is not a trained model. We do not claim the bucket brigade always beats fixed stations: with spare staff, a well-assigned fixed line is as good or slightly faster. Machine breakdowns, breaks and order mix are not modelled.

We make no claim about: **Singapore ownership structure** (a Chagee contact says outlets run under one company and one data system, and that the ownership form does not matter here; I-chagee-contact); **delivery profit by channel**; **Chagee's own equipment cost** (not public).

SPEAKER NOTES · SOURCES AND ASSUMPTIONS

Soft copy only.